

Intermittent Fasting and Sports Performance: A Review

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ABSTRACT

Intermittent fasting (IF) has been studied in athletes during Ramadan and in those willing to decrease adiposity while maintain or increasing lean body mass. The purpose of the systemic review was to summarize the effects of IF on sports performance outcomes. We searched peer-reviewed articles in the following databases: PubMed, Google Scholar, Google (up to 2019). Studies were selected if they included samples of adults, had experimental, or observational design. The twenty articles met the eligibility criteria. Finding indicated that maximum oxygen uptake is significantly enhanced, but reduced with intermittent fasting (body mass & fat mass). Non-significant effects were observed for muscle strength and anaerobic capacity. While IF may lead to impairments in aerobic capacity, as there are few studies per performance outcomes, more research is needed to move the field forward.

Keywords: *intermittent fasting, sports performance, aerobic capacity, anaerobic capacity, Ramadan fasting.*

INTRODUCTION

Intermittent fasting is a period of voluntary abstinence from food & drink has been practiced since earliest antiquity by people around world. The health outcomes of intermittent fasting are change in weight & metabolic parameters associated with type 2 diabetes, cardio vascular disease and cancer. Time restricted feeding is a form of intermittent fasting. Intermittent fasting recently gained popularity in ketogenic diet, weight loss & body building. It's an ancient practice. The science behind its many health benefits is just recently behind exposed to main stream society.

During fasting, basically allow our body to naturally cleanse, repair and regenerate itself for optimal function. three main health promoting mechanisms associated with fasting include the metabolic regulation of circadian biology, gut microbiome & different lifestyle behaviours. During the if feeding signals seem to be the main timing cue for how your circadian rhythms function & thus control certain metabolic, physiological & behavioural pathways that contribute to overall health & longevity.

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If can help synchronize our circadian rhythms leading to improved fluctuations in gene expression, reprogramming of energy metabolism & improved the hormonal & body weight regulation. IF has direct & positive influence on the gut microbiome through reduced gut permeability, diminished systemic inflammation, promotion of energy balance by enhancing gut integrity. Fasting puts an adaptive cellular stress on body which in turn allows your body to cope with more severe stressors that may occurs and thus protect against potential disease progressions.

Exercising during a fasting state increases lipolysis in adipose tissue while also stimulating peripheral fat oxidation resulting in increased fat utilization and weight loss. the most frequent form of dietary restriction is there are three types: 1) Intermittent calorie restriction/ fasting (ICR), 2) Alternate day fasting (ADF), 3) Time restricted feeding. Intermittent Caloric Restriction: it is known as whole day fasting & simplest form of fasting. It involves more than 24h for two or three times in a week.

Alternate Day Fasting: it is an intermittent feeding method. The basic idea is that the individual fast on one day & then eat whatever the individual wants the next daytime. Restricted Feeding: time restricted feeding involves a set number of hour daily fasting while feeding is permissible during the remaining hours. The rationale for TRF centers around Circadian rhythms. it

important of chrono -nutrition which is interaction between meal timing & the circadian system that regulates physiological cycles, metabolism & behavior.

Ramadan: It occurs during the month of Ramadan. the Ramadan is one of the five pillars of Islam. Ramadan is ninth month of the Islamic lunar calendar & daily abstinence from food & fluid intake occurs from dawn to sunset for duration of this month.

Impact of fasting: During fasting when foods are restricted & the stored glucose has been used up. then the body begins to burn fat as a source of energy which can result in weight loss & it can help to preserve muscle and reduce cholesterol levels.

Detoxification: In fasting state a process also occurs because any toxin stored in the body's fat are dissolved & removed from the body. Fasting can increase the level of endorphins feel good" hormones in blood which can have positive impact on mental well being. Blood glucose control: fasting promotes blood glucose control by reducing insulin resistance. decreasing insulin resistance can increase body's sensitivity to insulin allowing it to transport glucose from bloodstream to the cells more efficiently. Lowers the blood sugar preventing spikes and crashes of blood sugar level.

Fighting inflammation: The fasting could decrease several markers of inflammation & may useful in treating inflammatory conditions such as multiple

sclerosis. Boosting Brain functions and prevents neurodegenerative disorders: fasting improved both brain function & brain structure, fasting could protect brain health & increase the regeneration of nerve cells which enhance cognitive function.

Fasting can increase level of human growth building Fasting can increase level of human growth hormone (HGH) it is an important protein hormone that plays a important role in growth metabolism, weight less & muscle strength, muscle. While prolonged fasting harms health & physiological performance, it remains unclear whether shorter or more prolonged periods of intermittent fasting are necessarily harmful.

The effect of fasting on the performance of elite athletes started to be investigated in 2007, stimulated by the belief by athletes and coaches that fasting negative impacts on sports performance. In this review, we discuss the effects of intermittent fasting on body composition changes, metabolic adaptations and sports performance in trained and untrained subjects.

MATERIALS AND METHODS

The study assimilated variety of search method that are relevant to the content including scientific authentication. For this systemic review, a comprehensive and thorough search was conducted for procurement of evidence. It was also remember that while searching on open access that the articles

must be relevant to topic of research. In this particular study. The articles are major complementary to intermittent fasting and sports performance data bases.

The study related to objectives of paper were included the last few years reviews which are documented in English form the year 2002 -2019 were included the study that included intermittent fasting, sports performance, Ramadan fasting, endurance power, body composition, ketogenic diet, gene expression, specific sports related enhancement are part of this research. The studies which were not directly pointed towards the concept of analysis and the objective of research were excluded from the process.

Many scientific evidence indicate that intermittent fasting has a potential effect on weight loss, endurance power, and body composition. different types of electronic database that were already mention in above text, all were done in systemic manner.

Study Characteristics : The characteristics of all included studies are summarized in table. Half of the study are noncontrolled trials mainly aims to outcomes (aerobic, anaerobic and muscle strength adaptations) and change in body composition. This study included 10 women and 14 focused on the responses of trained men (football player, running athletes or resistance – trained men) The majority of IF protocols were conducted in laboratories and in gymnasiums or

training cramps.

More than half were undertaken during Ramadan and others explored the effects of IF vs normal diet during a period of 8 weeks, 6 weeks of overnight fast and 3 to 10 days. participants are age group between 18-39 years. This study included women, healthy and moderately active samples, six studies tested sedentary and untrained men, 10 focused on the responses of trained men (soccer player, running athletes or untrained men).

Outcomes :

Change in muscle - Twelve studies presented data for association between IF and muscle with approximately 18% of studies observing significant changes. The upper arm girth and maximum voluntary contraction was improved during IF.

Change in aerobic capacity - Fifteen studies reported adaptation in aerobic capacity with approximately 36% of attaining significant changes in specific outcomes. Some studies obtained significant decrease in VO₂max, heart rate, VCO₂. After if fasting or Ramadan performance like: sprint performance, running speed, physical activity was reduced. (Vicente, S. et al, 2015), (Chaouachi, A. et al, 2009), (Santos, H. et al, 2018).

Change in Anaerobic capacity - Only seven studies reported results for anaerobic changes with approximately 11% showing negative but effect of intermittent fasting on maximal speed

during high intensity effort. but 7% of studies shows no effect of IF in anaerobic capacity. (Vicente, S. et al, 2015)

Changes in body composition - Body composition are frequently studied and approx. 68% of studies reported that IF impact on body weight loss, it also reduced the blood volume, blood glucose level, reduced muscle glycogen store, reduced liver glycogen store, reduced body fat %, reduced fat mass. (Antoni, R. et al, 2017).

Changes in blood parameters - Sixteen studies shown that BHB (beta hydroxybutyrate), PGC-1 alpha, ATGL (adipose triglyceride lipase), AMPK (AMP-activated protein kinase), PEPCK (phosphoenol pyruvate factor), Tumor necrosis factor- alpha (TNF-alpha), adiponectin, leptin, ghrelin, IL6, CRP (c-reactive protein) are increases during the IF which is associated with body weight loss. ((Levy, E., & Chu, T., 2019), (Mattson, M. P. et al, 2018).

The five studies shows that blood cholesterol, corpuscular hemoglobin concentration, blood total cholesterol, blood glucose level, plasma insulin level, liver glycogen store, testosterone, TSH (triiodothyronine), IMCT (intra myocellular triglycerides), BUN (blood urea nitrogen), ALT (alanine transaminase), GH (growth hormone), CK (creatine kinase), LDH (lactate dehydrogenase) are all decreased during intermittent fasting, CK is a key molecule of Pcr (phosphocreatine) which is the main source of high intensity short

duration activity due to decrease in CK IF is not beneficial .(Santos, H. et al, 2018),(Urooj, A. et al, 2020).

(Chaouachi et al.2009) all observed no decrease in vertical jump, squat jump or counter-movement jump scores during Ramadan, although the average power of the judokas during 30 s of repeated jumps did decrease slightly, from 23.4 to 22.4 W/kg. The repeated jumping was final test of the day, when fluid depletion would have been maximal, although the decrement of performance could also reflect a decrease of central drive associated with a fall of blood glucose levels.

Change in cerebral function - Information on mood changes during Ramadan has commonly been obtained from the Profile of Mood States (POMS) questionnaire; where change has been observed, there has been an increase in fatigue scores. Chaouachi and colleagues studied the effects of a November Ramadan observance in 8 middle-distance athletes. Their sample showed a trend towards a decrease of body mass and body fat, with a significant 8% decrease of energy intake by the 21st day of Ramadan. Sleep was also substantially curtailed (by 20% on day 21). The POMS fatigue score was increased by Day 21 of Ramadan, but there were no significant changes in scores for the other 5 POMS scales.(Chaouachi, A. et al,2009).

DISCUSSION

The aim of comprehensive systematic review was to summarize the

effects of two approaches of IF on performance outcomes through adaptation in muscle strength, aerobic, anerobic capacity; as well as the body composition. Twenty studies were included that intermittent fasting affects the VO2max specifically.

During Ramadan, IF aerobic capacity was shown to be affected with a significant decrease in VO2max. one possible explanation for these findings is that the dehydration accompanying Ramadan might reduce blood volume, maximal cardiac output and muscle glycogen stores, thus affecting VO2max. However, it is important that aerobic capacity had verified in early phases of Ramadan IF that reduced.(Chaouachi, A. et al,2009).

Many studies show that TRF exerts a positive impact on VO2max. a possible explanation for improvement of VO2max with TRF includes an increase of cardiac output due to increase of sympathetic stimulation during exercise and a greater oxidative capacity in fat adapted muscle.(Alam, I. et al.,2019) Ultimately, the combination of these phenomena may lead to improvements in aerobic performance in EU hydrated condition, thus training in fasted state could provide an adjuvant stimulus to enhance training adaptations while preserving the plasma concentration at a normal level during physical activity.

Improvement of VO2 max is a partially related to positive adaptation in body composition, fasting triggers

numerous hormonal adaptations culminating in enhance metabolic rate and preserved lean body mass, increase the level of noradrenaline and growth hormone.

Fasting increases the expression of sirtuin1 (SIRT1) and phosphorylate AMP- activated protein kinase (AMPK) which have numerous effects on gene expressions, regulation of mitochondrial biogenesis. (Levy, E., & Chu, T.,2019).

Although past research showed that IF dose not negatively effect on muscle damage, other reports observed small decrease in muscle strength. It was argued that such negative effects arise slightly due to the reductions of fat mass, dehydration or alterations in energy metabolism, muscle buffering capacity during the muscle contractions and different lengths of performance.

There is compelling evidence that the ingestion of selected amino acids or metabolites during IF may be effective for maintenance of lean body mass. It has been argued that leucine, a branched chain amino acids play a pivotal role in stimulating protein synthesis through the mTOR signaling cascade, thus modulating these physiological effects.(Ahmed,A.etal,2018).

During the fasting fat are burn as a source of energy which can help in toxin remove from the body. It causes detoxification. Fasting could decrease several markers of inflammation and may decrease the risk of several inflammatory conditions.

The hormonal endpoint, cortisol which is a metabolic and circadian hormone, and BNDF is increases during IF and it promotes neuronal growth. During IF we additionally expected to observed a decrease in IGF-1 which are associated with cancer risk and aging(Chaouachi, A. et al,2009), (Patterson, R. E., & Sears, D. D.,2017). Further research is needed to explore the interaction between IF & exercise performance in women sports person. It is a fundamental to ensure that both volume and intensity of training are well maintained during experiments. Nutrient intake should be closely monitored over the time and it is mandatory to determine the role of alter sleep anddehydration on relationship between IF and exercise performance. Finally, more studies related with the effects of IF on performance are also required to unravel the physiological mechanisms and energy pathways that allow the athletes to maintain or even improve performance during IF.

CONCLUSION

Overall, the existent research examining the effects of IF on physical parameters and body composition has provided conflicting data. Additionally, they suggest that IF may also positively effects on decreasing body mass and fat mass.

For muscle strength and anaerobic capacity, the pooled estimated effects of

intermittent fasting were largely no significant, thus the current review indicates that's while Ramadan IF may lead to impairments in aerobic capacity, TRF may be effective for improving it. Intermittent fasting is a promising strategy among different approaches of fasting such as caloric restriction, dietary

restriction and intermittent fasting. Intermittent fasting is a most fruitful approach for its ability to cope up with different diseases which resulting in prolong lifespan, help in aerobic metabolism and improved in endurance activity.

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